॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
Indian Institute of Technology Dharwad

# CS101C: Introduction to Programming (Using C)

**Instructors:** Prof. Aikta Arya and Prof. Konjengbam Anand

CSE Department

Indian Institute of Technology, Dharwad

August 3, 2026

# CS101C: Computer Programming



1. **Description:**
   - This course provides an introduction to problem solving with computers using 'C' as a programming language.
2. **Credit structure (L-T-P-C) : 3-0-3-9**
   - 3 contact hours (three 50min lectures) per week. 6 credits.
   - 3 lab hours (150 mins in total) per week. 3 credits.
   - Full-semester (14 week + 2 exam week) core course.
3. **Prerequisites: None**

1. **Assessment Plan - Theory**
   - Two paper based exams: Midsem and Endsem
   - Weightage: Midsem = 40 points, Endsem = 60 points
2. **Assessment Plan - Lab**
   - Lab 1 - Practice.
   - Labs 2-3 = 6 points each (12 points in total)
   - Labs 4-14 = 8 points each (88 points in total)
3. **Grades:**

| If your numerical score is at least: | Your course grade will be at least: |
|---|---|
| 90 | AA |
| 80 | AB |
| 70 | BB |
| 60 | BC |
| 50 | CC |
| 45 | CD |
| 40 | DD |

# CS101C: Computer Programming



1. **Developer essentials:**
   - Editors, Integrated Development Environment (IDE), Unix Shell, Library-based development, Compiler toolchain
2. **Programming in C**
   - Machine representation, data types and control flow, operators, arrays and strings, functions and recursion, pointers and structures, Input and output using files.
3. **Applications:**
   - Sample problems in engineering, science, text processing, and numerical methods.

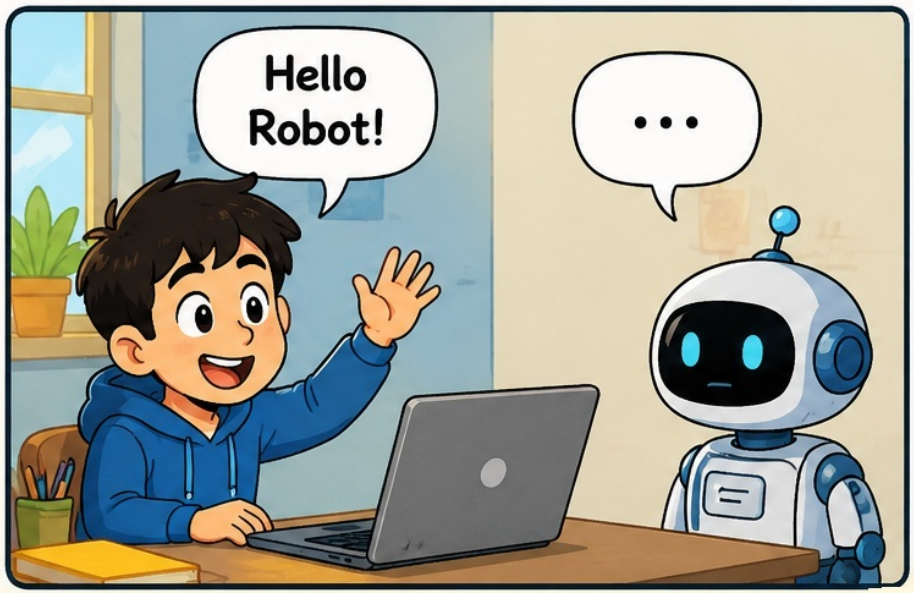
॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

## 1 References and Texts:

1. **The C Programming Language,** Brian W Kernighan, Dennis M Ritchie, Prentice Hall India , 2nd edition, 1988
2. **Programming with C (Second Edition)** Byron Gottfried, Schaum's Outlines Series, Tata-Mcgraw Hill, 2011
3. **How to Solve It by Computer,** by G. Dromey, Prentice- Hall, Inc., Upper Saddle River, NJ, 1982.
4. **How to Solve _It (2nd ed.),** by Polya, G., Doubleday and co, 1957.
5. **Let Us C**, by Yashwant Kanetkar, Allied Publishers, 1998.
6. **Programming in ANSI C,** by E. Balaguruswamy

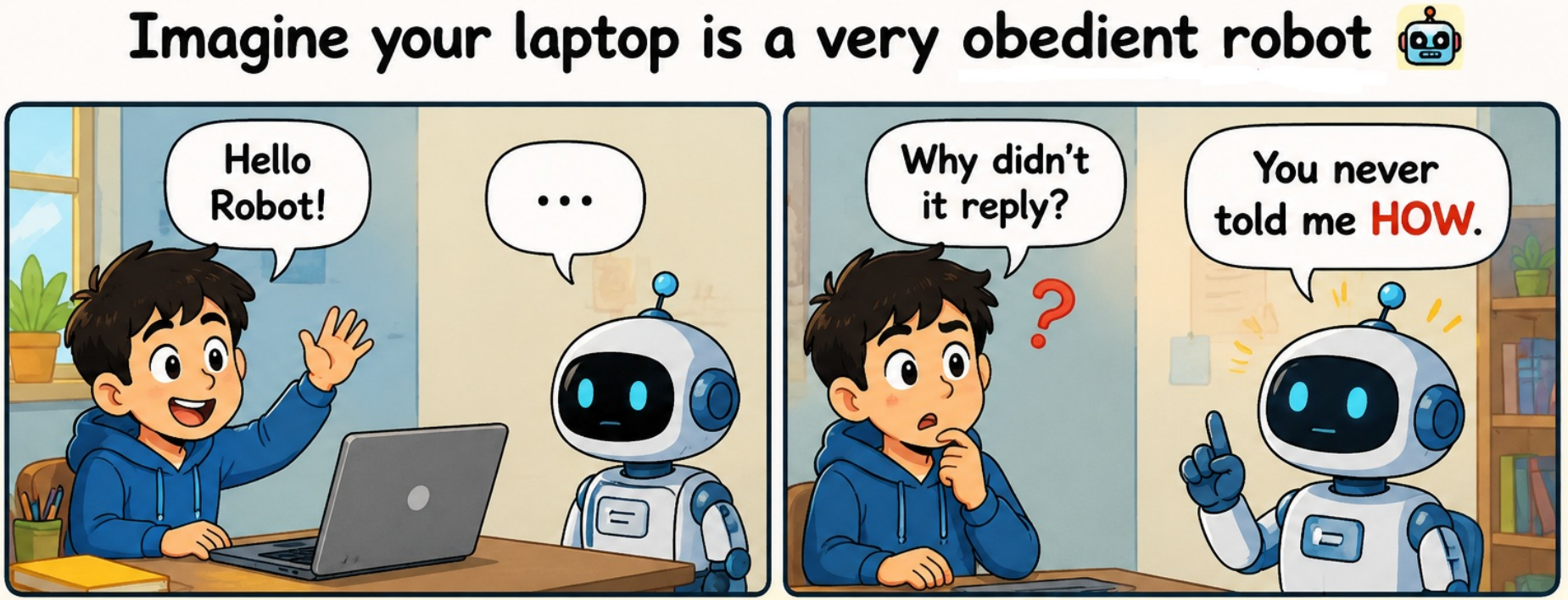
# Why Programming?

## Imagine your laptop is a very obedient robot



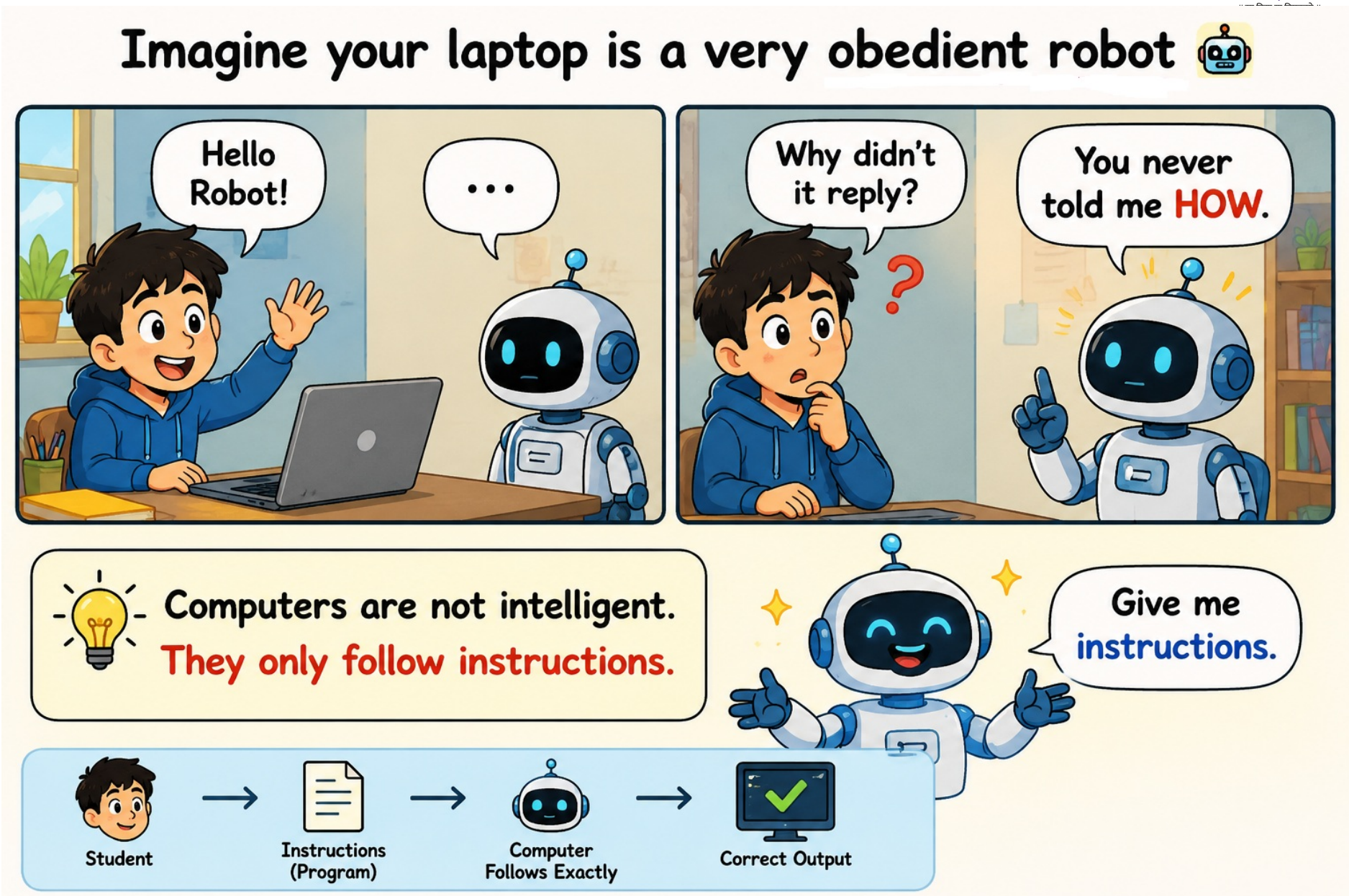

[1] Courtesy: OpenAI (generated using ChatGPT)

# Why Programming?




[1] Courtesy: OpenAI (generated using ChatGPT)

# Why Programming?



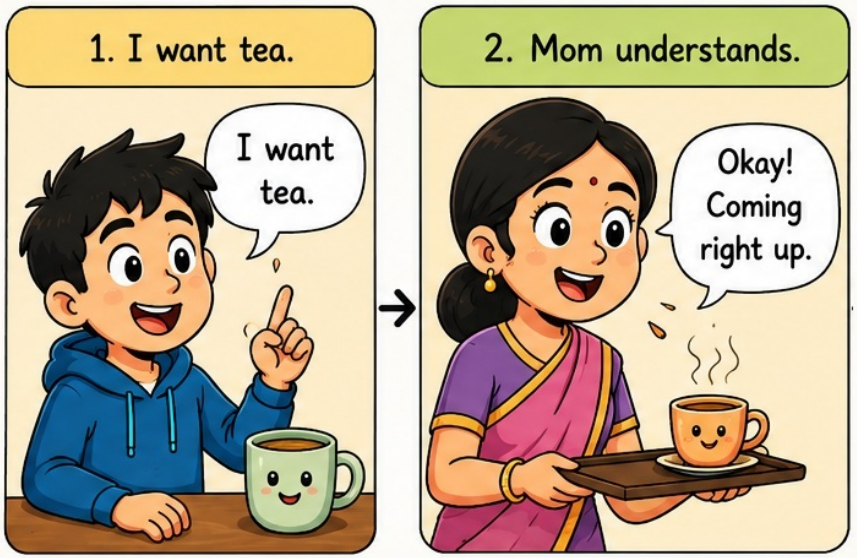

[1] Courtesy: OpenAI (generated using ChatGPT)

# What is Programming?



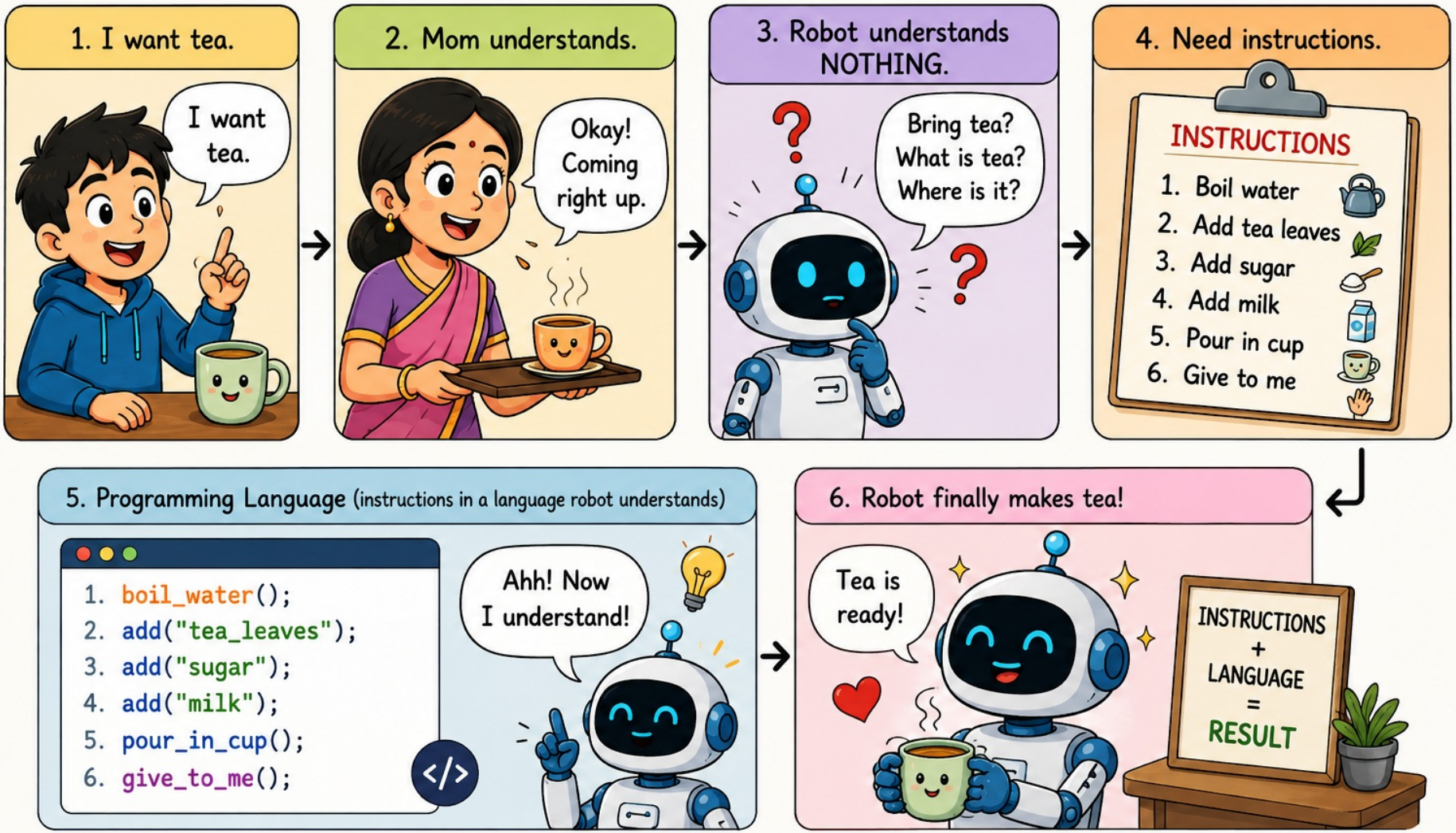

[1] Courtesy: OpenAI (generated using ChatGPT)

# What is Programming?




[1] Courtesy: OpenAI (generated using ChatGPT)

# What is Programming?




[1] Courtesy: OpenAI (generated using ChatGPT)

# Humans Vs Computers

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad



## Human



## Computer





### Intelligent

Humans can think, learn and make decisions.



### Extremely obedient

Computers do exactly what they are told. Nothing more, nothing less.





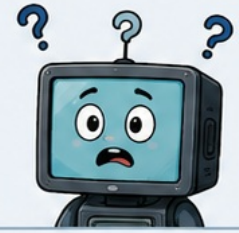

### Can guess

Humans can make good guesses even with incomplete information.



### Never guesses

Computers never guess. They only do what they are explicitly told.






### Understands mistakes

Humans understand mistakes and can forgive.



### Doesn't forgive mistakes

Computers show errors and stop if something is wrong.





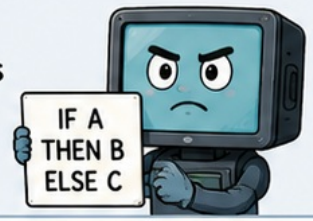

### Flexible

Humans adapt to new situations and are flexible.



### Extremely literal

Computers follow rules literally. They don't adapt unless reprogrammed.



# Computers need precise instructions

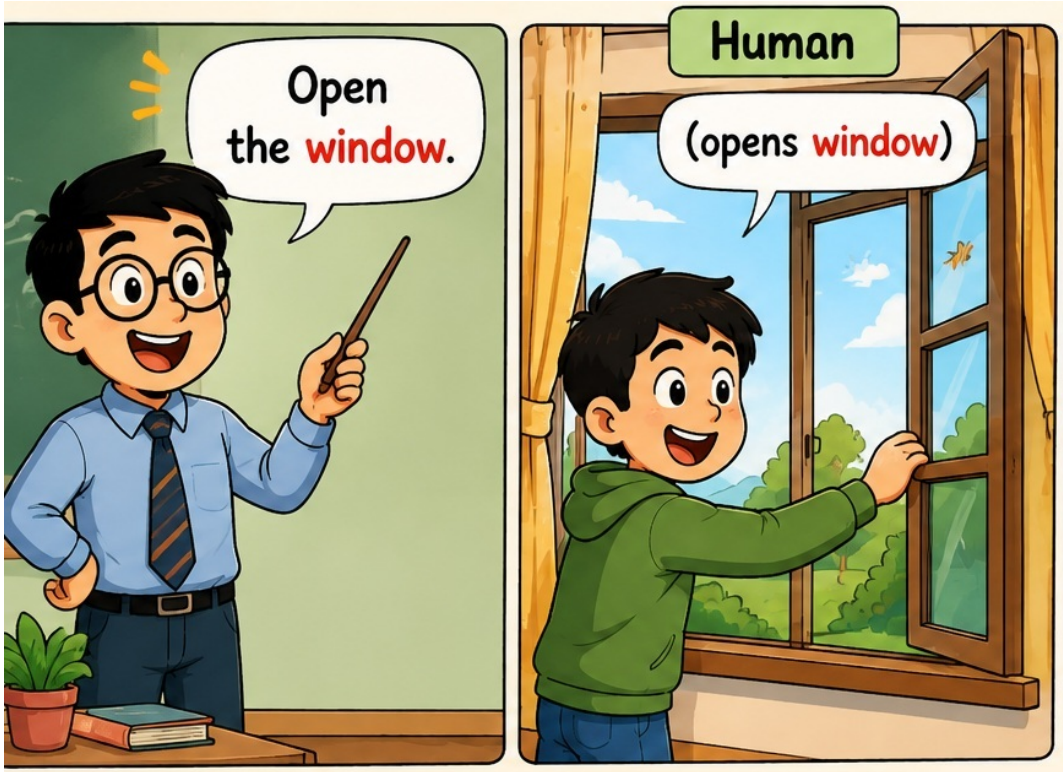
॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad




[1] Courtesy: OpenAI (generated using ChatGPT)

# Computers need precise instructions

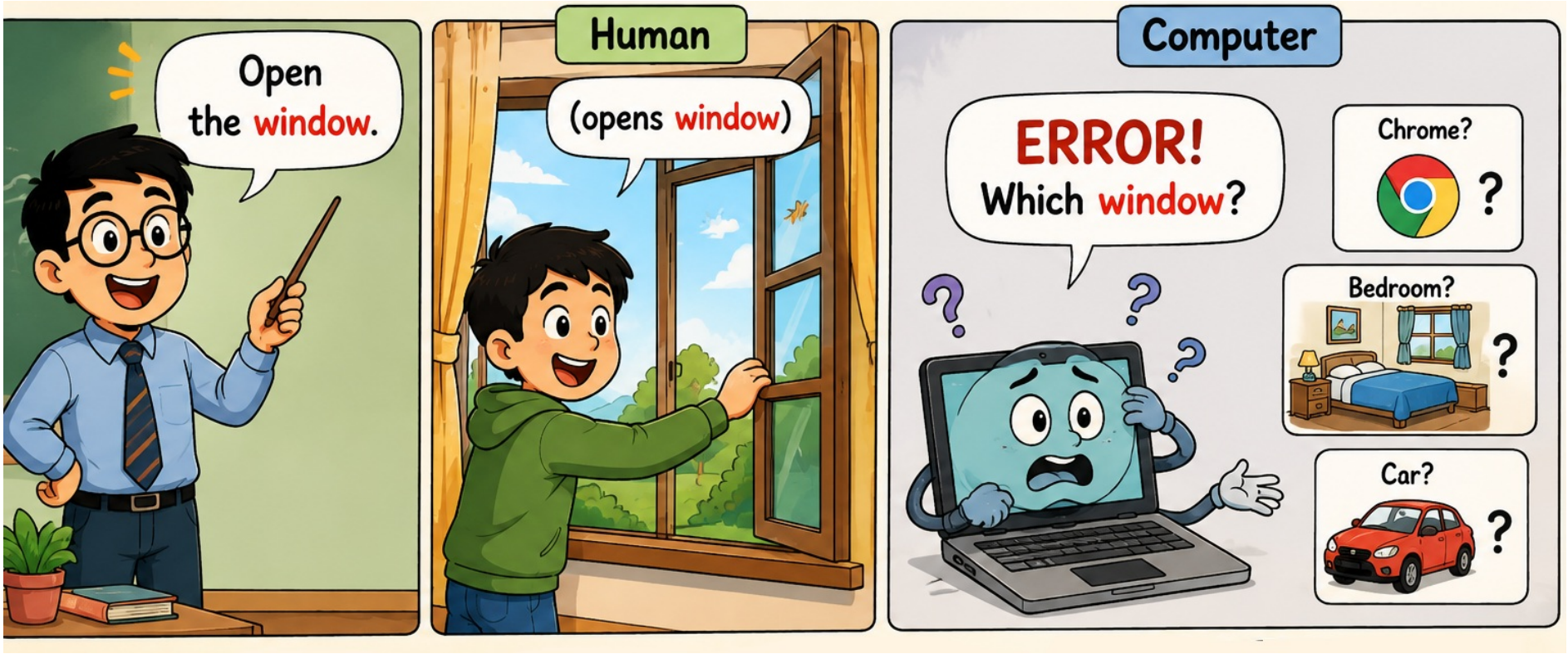





[1] Courtesy: OpenAI (generated using ChatGPT)

# Computers need precise instructions

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad




[1] Courtesy: OpenAI (generated using ChatGPT)

# Full Computer System





**Monitor**
Shows the output (what we see).

**Keyboard**
Used to type and give input.

**Mouse**
Used to move the pointer, select and click.

**Microphone**
Takes sound as input.

**Speakers**
Gives sound (output).

**CPU**
The Brain of the computer. It thinks and processes everything.

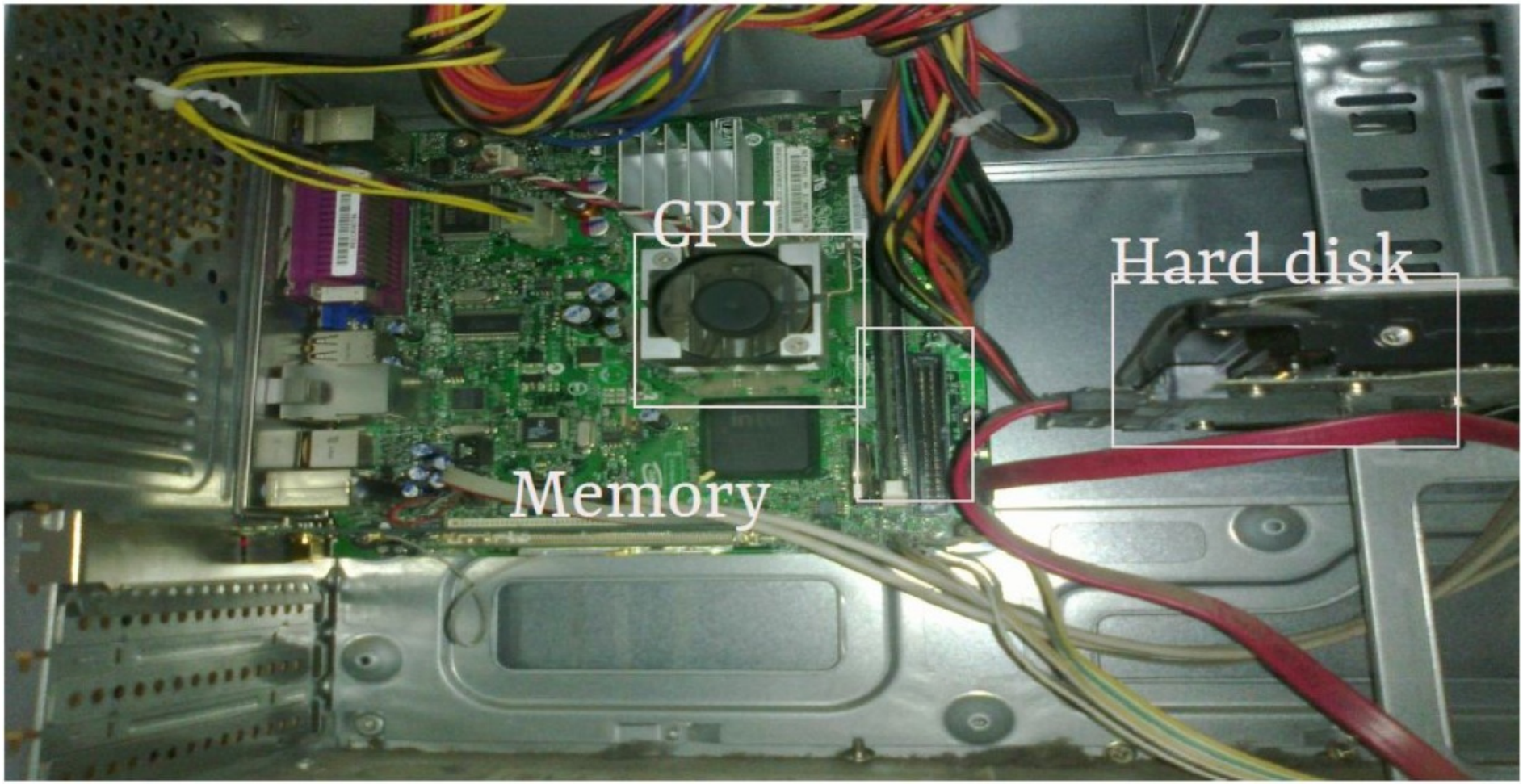
**RAM (Memory)**
Short-term memory. Stores data temporarily while computer is working.

**Hard Disk (Storage)**
Long-term storage. Saves all your files, photos, videos, programs, etc.

**Printer**
Prints digital content on paper (hard copy).

# What does a computer looks like?





# What is C?



- Dennis Ritchie is the inventor (creator) of the C programming language.
- He developed C in 1972 while working at **AT&T Bell Laboratories**.
- When Dennis Ritchie invented C, he wrote it for a computer called the **16 bit DEC PDP-11.**
- Widely used today.
- **Extends to newer system architectures:** Intel processors, AMD processors, ARM processors (mobile phones) Raspberry Pi, Supercomputers.
- **Efficiency/performance:** C programs execute very fast.
- **low-level access:** Many programming languages keep programmers far away from the hardware, unlike C, which allows you to work very close to the hardware.

# What is C?

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
Indian Institute of Technology Dharwad



Figure: Dennis Ritchie (standing) with Ken Thompson at work in 1972 in front of a DEC PDP-11.

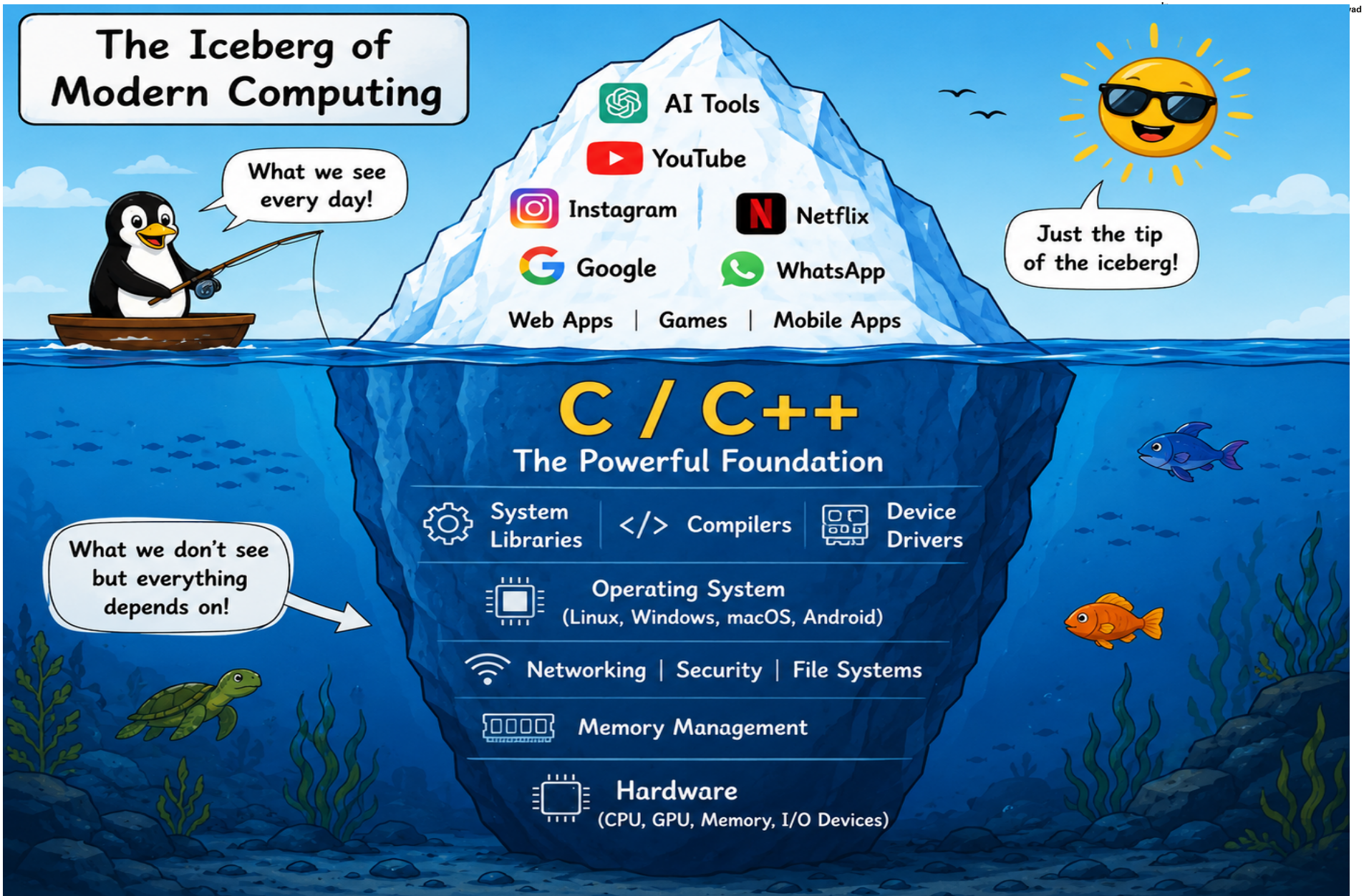
# Is C still relevant in the AI Era???



# Is C still relevant in the AI Era???

॥ सा विद्या या विमुक्तये ॥

भारतीय प्रौद्योगिकी संस्थान धारवाड



# Is C still relevant in the AI Era???

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड

## Operating Systems

Windows, Linux, macOS,
Android, iOS (core parts)
All start with C!



C code runs
before you even
see the desktop!

## Embedded Systems

Cars, Airplanes, Satellites, Rockets,
Medical devices, Smart watches,
IoT devices and many more!




C is used where performance, reliability
and low-level control are critical.

## Internet & Networking

Web servers, browsers,
networking software,
routers – mostly in C/C++.



Every time you browse, stream
or connect – C is working behind
the scenes!

## Banking & Finance

ATMs, Card readers, POS machines,
trading systems – many use C.




Security, speed and reliability
are critical in financial systems.

## Compilers

GCC, Clang, and many tools
that compile your code are
written in C/C++.




Even the tools that build your code,
are built using C!

## Robotics & Automation

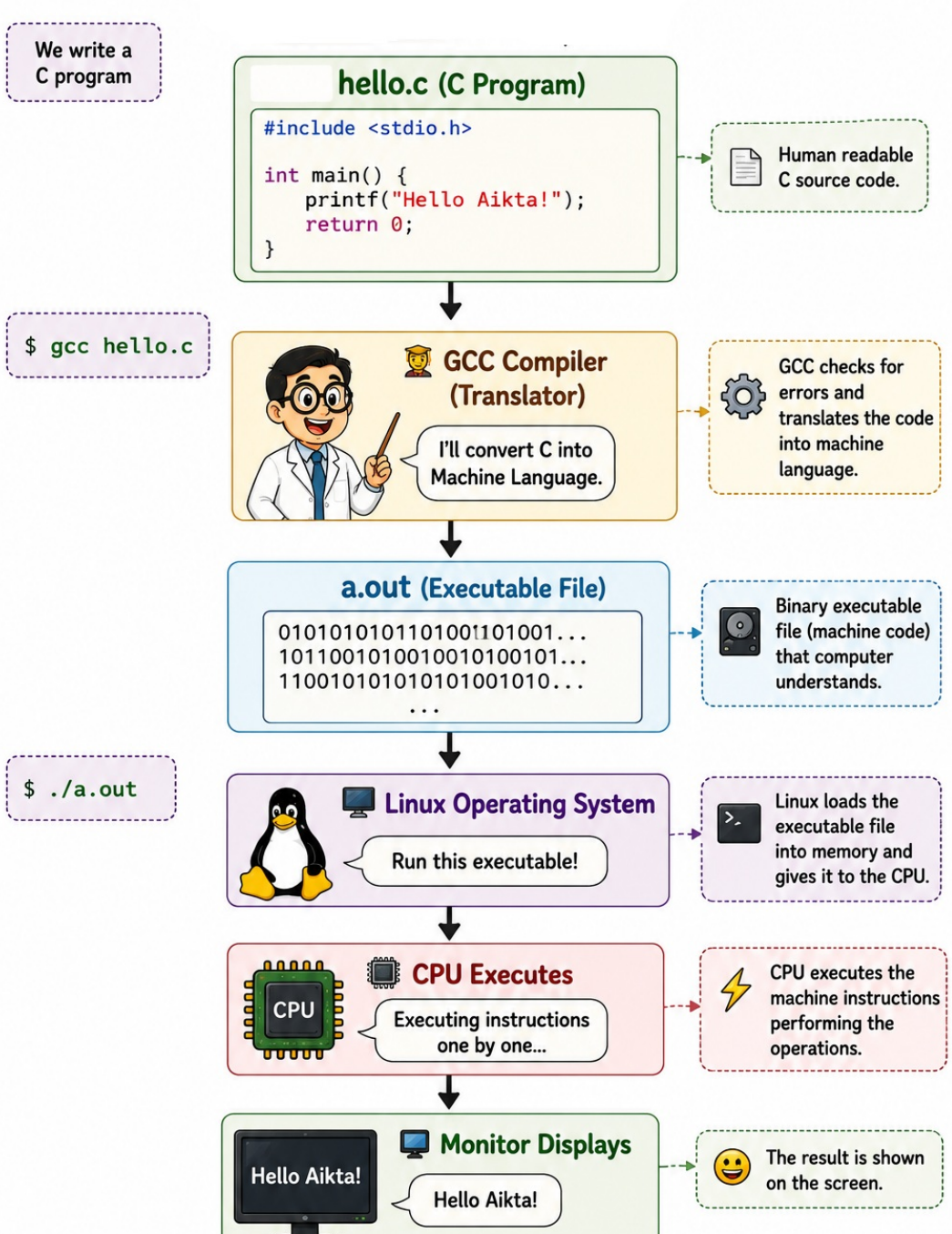
Robots, Drones, Industrial machines –
real-time systems use C/C++.



C gives the speed and control needed
for real-time and embedded systems.

# Writing C Programs





# Understanding printf()




You write this line of code

**printf:** instruction to print on the screen

```
printf("Hello World");
```




Output on the screen

| printf | ( | "Hello World" | ) | ; |
|---|---|---|---|---|
| **Function (Verb)** | **Start of function** | **Argument (Object/Data)** | **End of function** | **End of statement** |
| Tells the computer what action to perform. | Marks the beginning of the information (inside). | The data on which the function acts. Here, a string (sequence of characters) to be printed. H e l l o W o r l d Each box is a character | Marks the end of the information (inside). | Tells the compiler that this instruction is finished. |
| Action: Print | Start | | End | Finish |

### In English vs In C

| English | C Programming |
|---|---|
| Speak "Hello World" | printf "Hello World" |

- Argument is enclosed in quotes " "
- Strings are used to store text (names, urls, paragraphs, etc.)

More on strings later!

# Understanding printf()

॥ सा विद्या या विमुक्तये ॥




You write this line of code

**printf:** instruction to print on the screen

```
printf("Hello World");
```




Output on the screen

| printf | ( | "Hello World" | ) | ; |
|---|---|---|---|---|
| **Function (Verb)** | **Start of function** | **Argument (Object/Data)** | **End of function** | **End of statement** |
| Tells the computer what action to perform. | Marks the beginning of the information (inside). | The data on which the function acts. Here, a string (sequence of characters) to be printed. H e l l o W o r l d Each box is a character | Marks the end of the information (inside). | Tells the compiler that this instruction is finished. |
| Action: Print | Start | | End | Finish |

### In English vs In C

| English | C Programming |
|---|---|
| Speak “Hello World” | printf “Hello World” |

- Argument is enclosed in quotes “ ”
- Strings are used to store text (names, urls, paragraphs, etc.)
- More on strings later!

⭐ **Next:** More useful stuff – moving from a line of code to multiple lines!



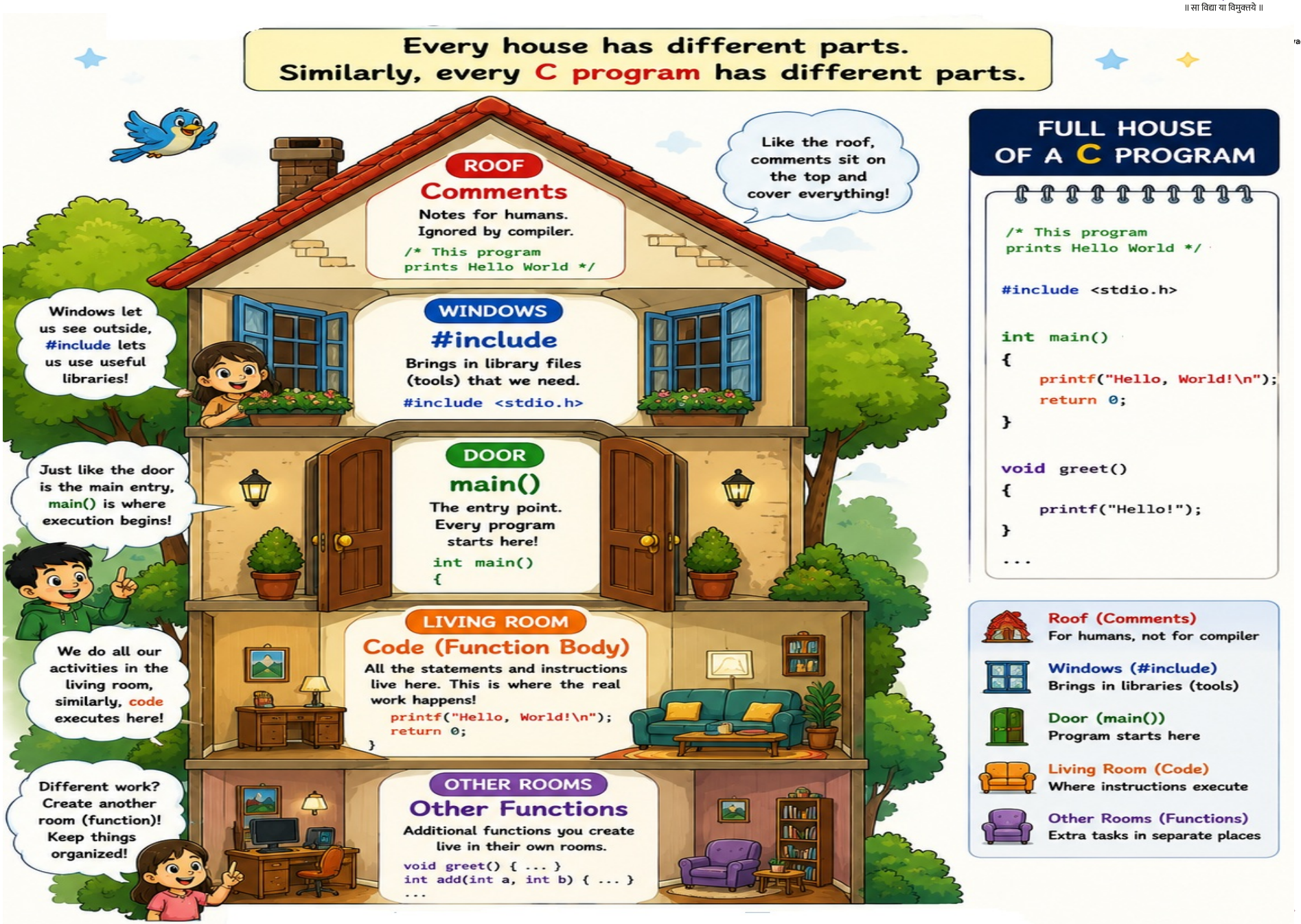



# Our First C Program!!

# Today's Class (5/8/2026)

# Anatomy of a C Program

॥ सा विद्या या विमुक्तये ॥



# Structure of a .c file

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड

```
/* Begin with comments about file contents */

Insert #include statements and preprocessor
definitions

Function prototypes and variable declarations

Define main() function
{
  Function body
}

Define other function
{
  Function body
}
⋮
```

# Comments: /* I am Comment */

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

## COMMENTS in C → For Humans, Not for Computers!

Think Like a Human!

Comments are not executed.
They are notes for humans who read the code.

Compiler says: "I don't care!"

### 1. Imagine a Notebook

Before doing an experiment, you write important details in your notebook.




### 2. Same in C Program!

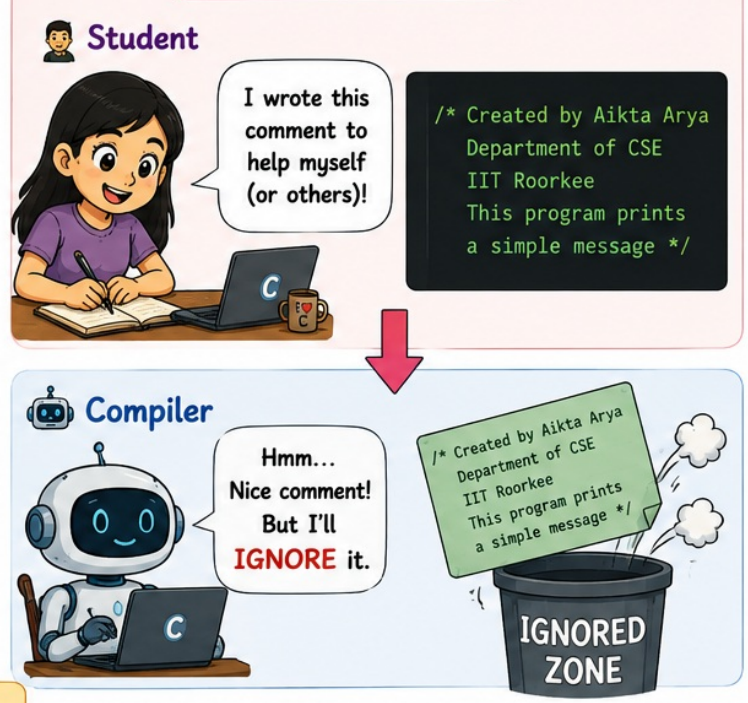
Comments are like notes for the programmer.

```c
/* Created by Aikta Arya
   Department of CSE
   IIT Roorkee
   This program prints
   a simple message */

#include <stdio.h>

int main()
{
   printf("Hello World!\n");
   return 0;
}
```

### 3. What Happens?




**BOTTOM LINE:** Comments don't change the output. They make the code easy to understand!

# Comments: /* I am Comment */

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

## COMMENTS in C → For Humans, Not for Computers!

Think Like a Human!

Comments are not executed.
They are notes for humans who read the code.

Compiler says: "I don't care!"

### 1. Imagine a Notebook

Before doing an experiment, you write important details in your notebook.




### 2. Same in C Program!

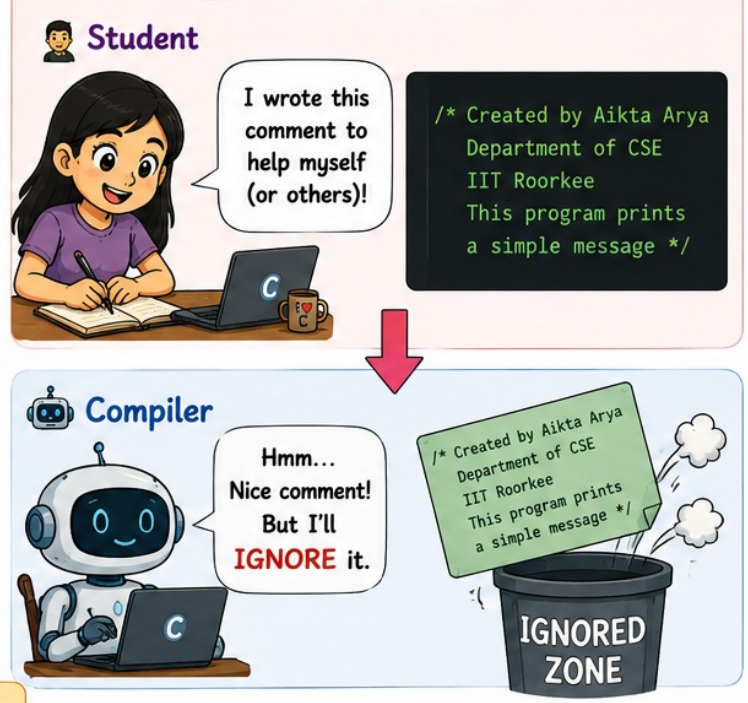
Comments are like notes for the programmer.

```c
/* Created by Aikta Arya
   Department of CSE
   IIT Roorkee
   This program prints
   a simple message */

#include <stdio.h>

int main()
{
   printf("Hello World!\n");
   return 0;
}
```

### 3. What Happens?

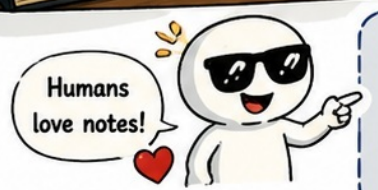



**BOTTOM LINE:** Comments don't change the output. They make the code easy to understand!

# Comments: /* I am Comment */

॥ सा विद्या या विमुक्तये ॥
धारवाड
y Dharwad



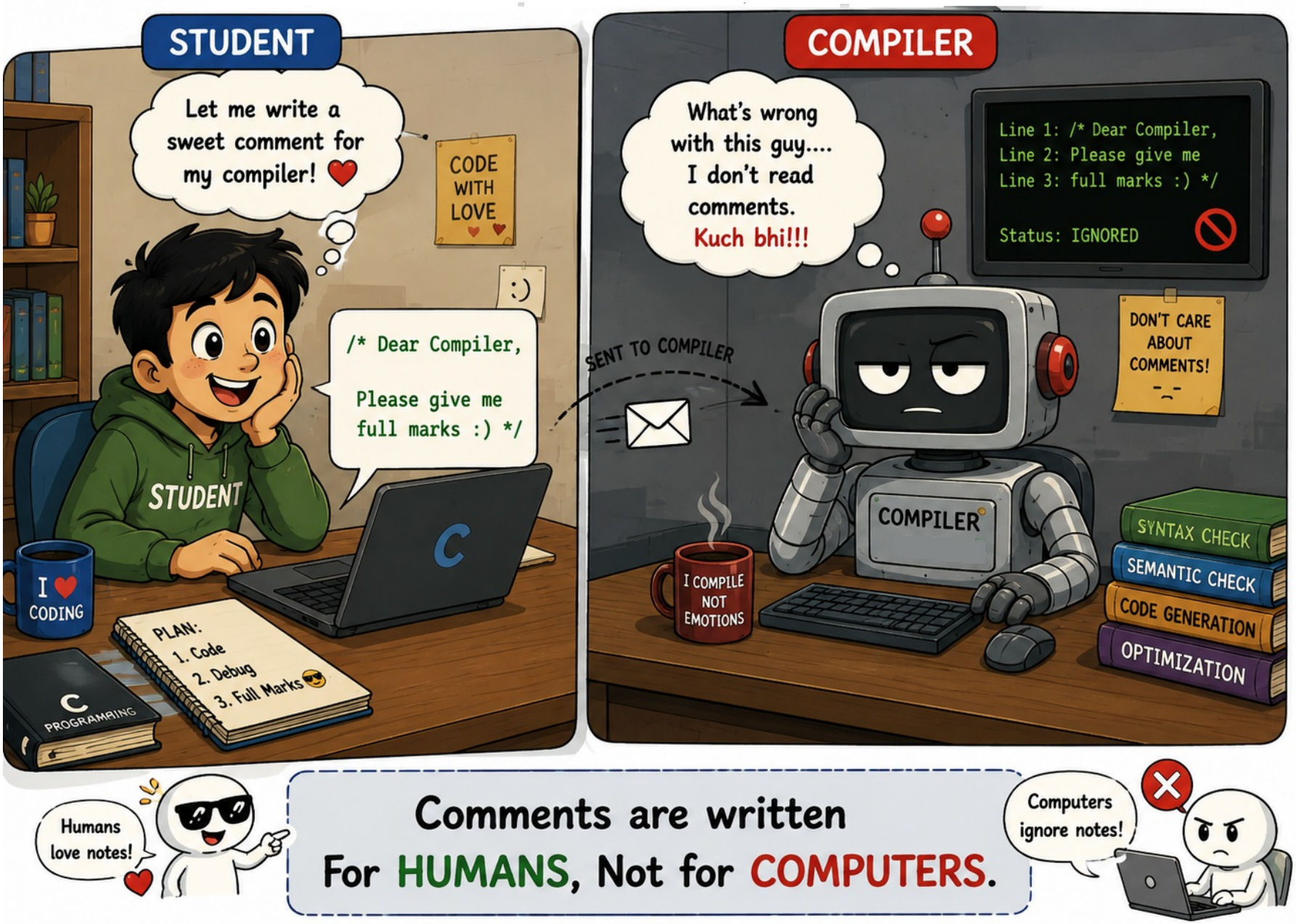




Comments are written
For HUMANS, Not for COMPUTERS.



# Comments: /* I am Comment */

॥ सा विद्या या विमुक्तये ॥
y Dharwad



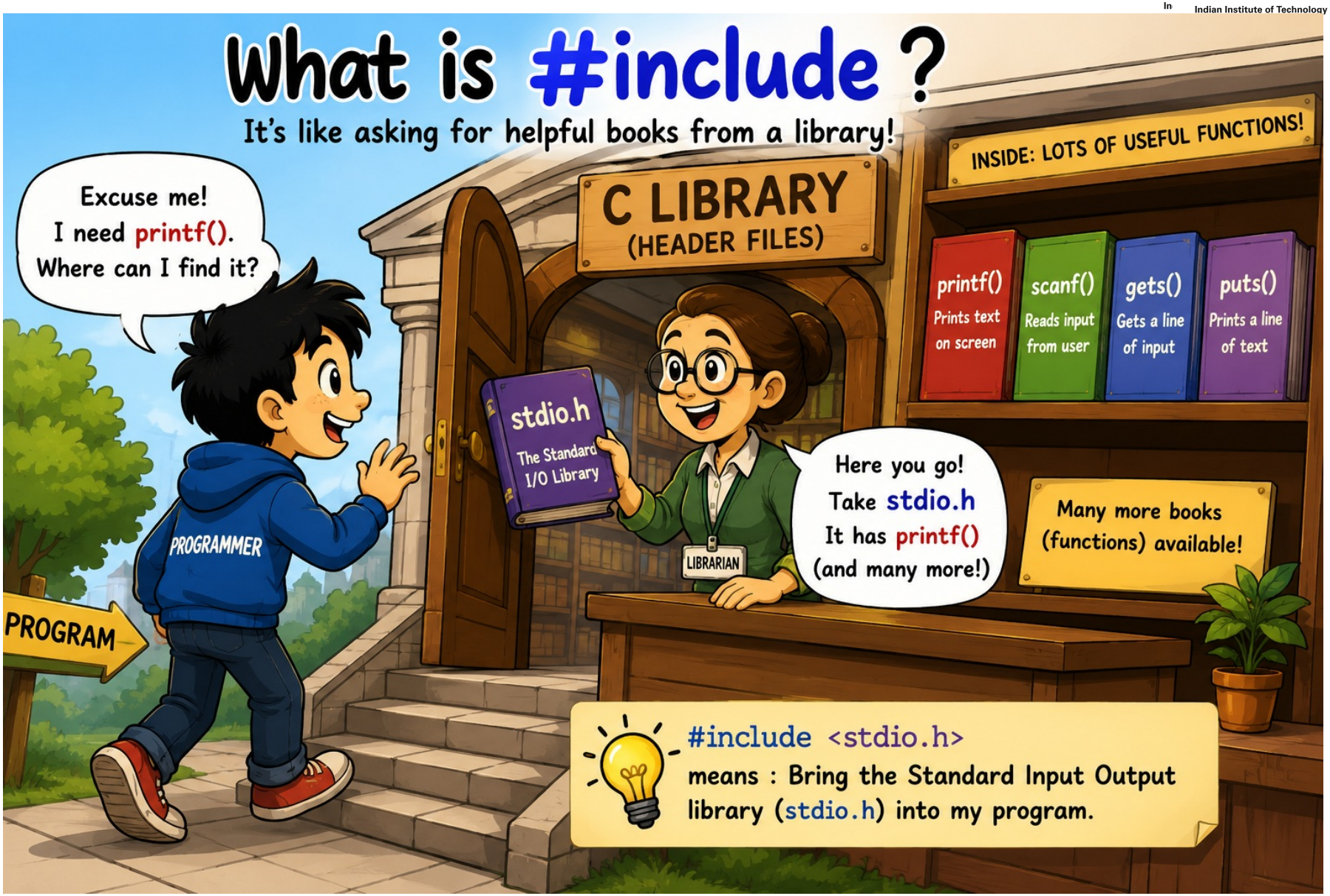
# Comments: /* I am Comment */

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

- Comments: /* this is a simple comment */
- Can span multiple lines: /* This comment
  spans
  multiplelines */
- Completely ignored by compiler.
- Can appear almost anywhere:
  /* hello.c – our first C program
  Created by Aikta Arya , 05/08/2026 */

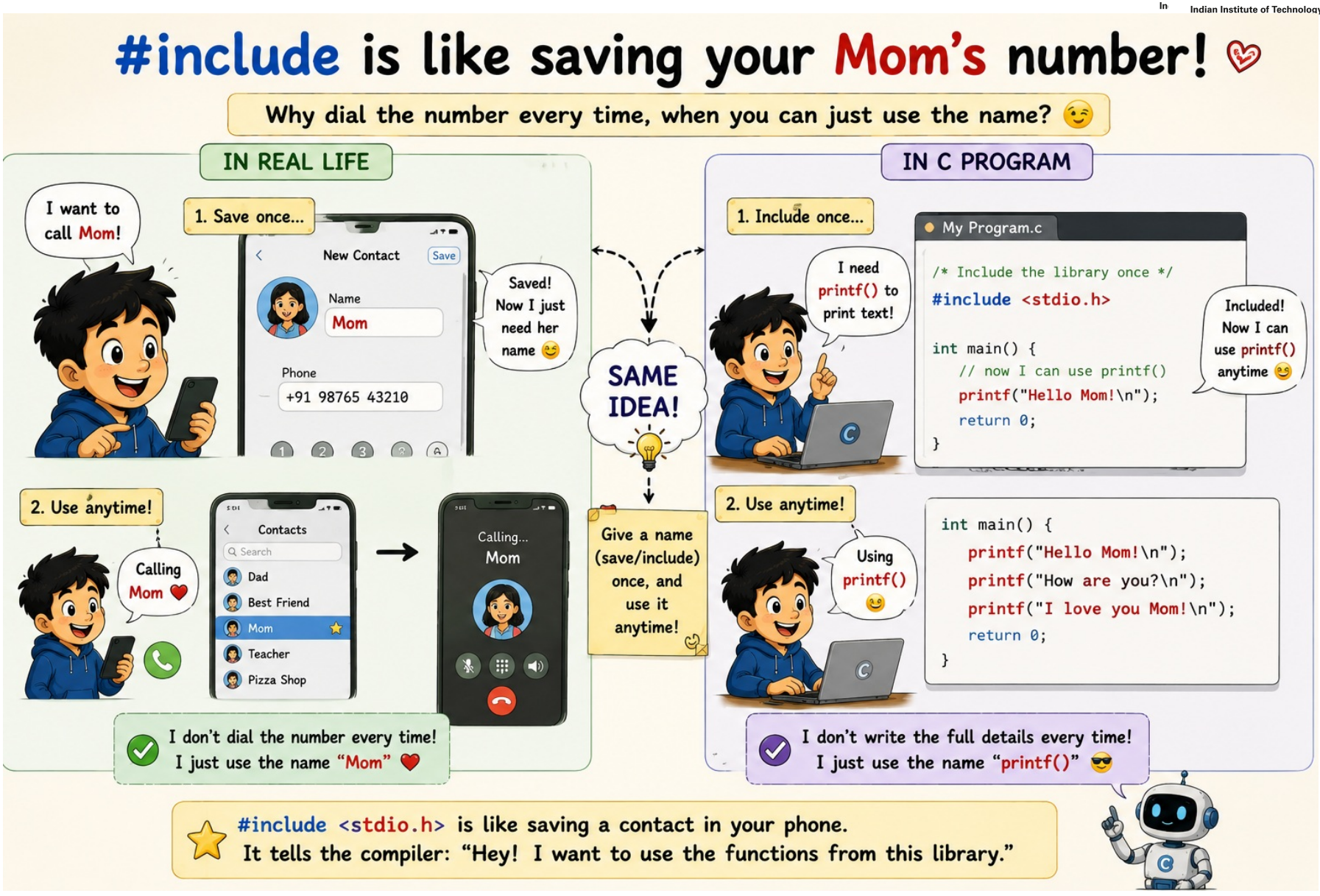
# The #include macro





# The #include macro

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad



# The #include macro

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

- Header files: constants, functions, other declarations
- #include < stdio.h > – read the contents of the header file stdio.h
- stdio.h: standard I/O functions for console, files:

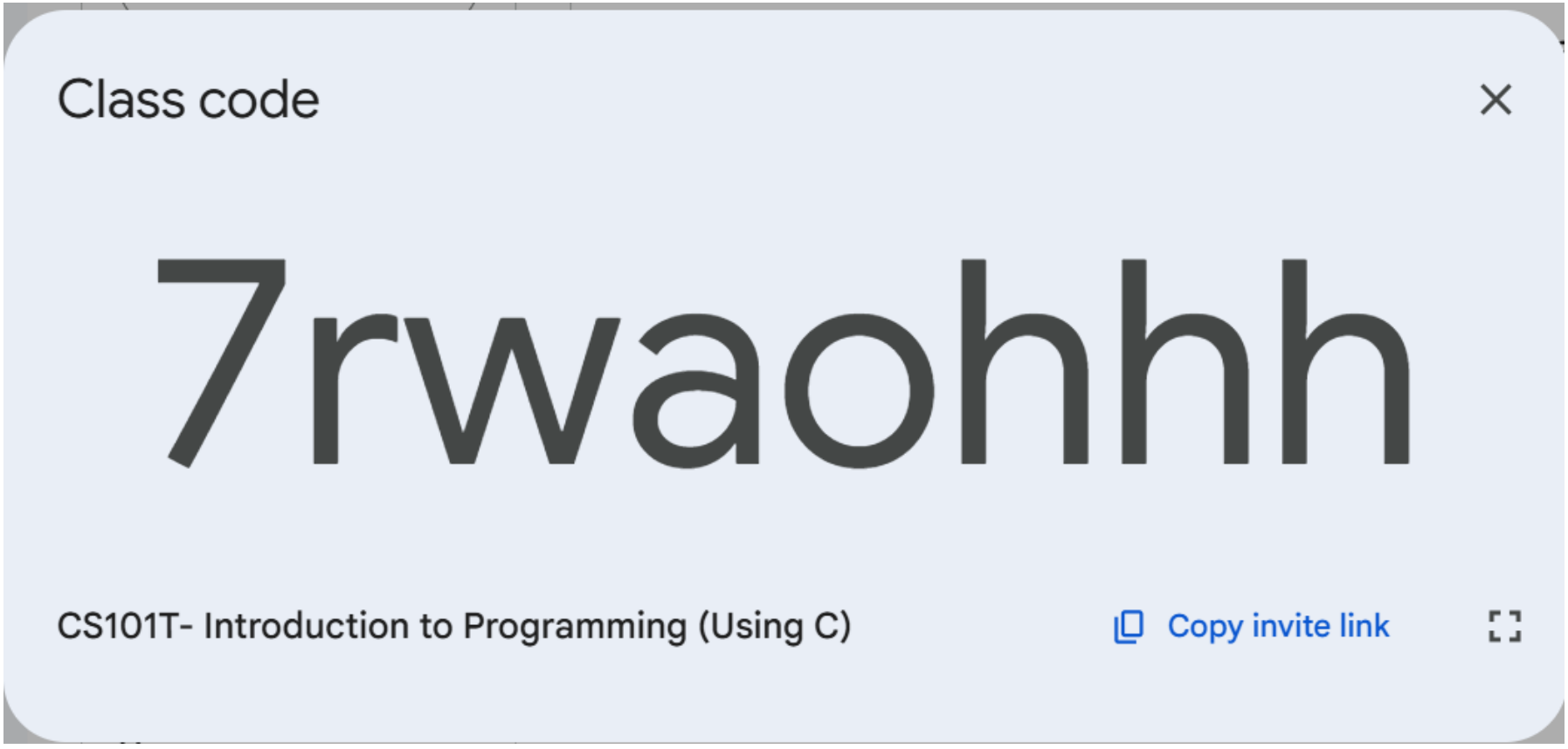
```
/* h e l l o . c – our first C program
Created by Aikta Arya , 05/08/2026*/
#include < stdio.h > /* basic I/O facilities */
```

- < stdio.h > – part of the C Standard Library
- other important header files: math.h, stdlib.h, string.h, time.h
- #include < stdio.h > – searches ./ for stdio.h first

# Today's Class (7/8/2026)

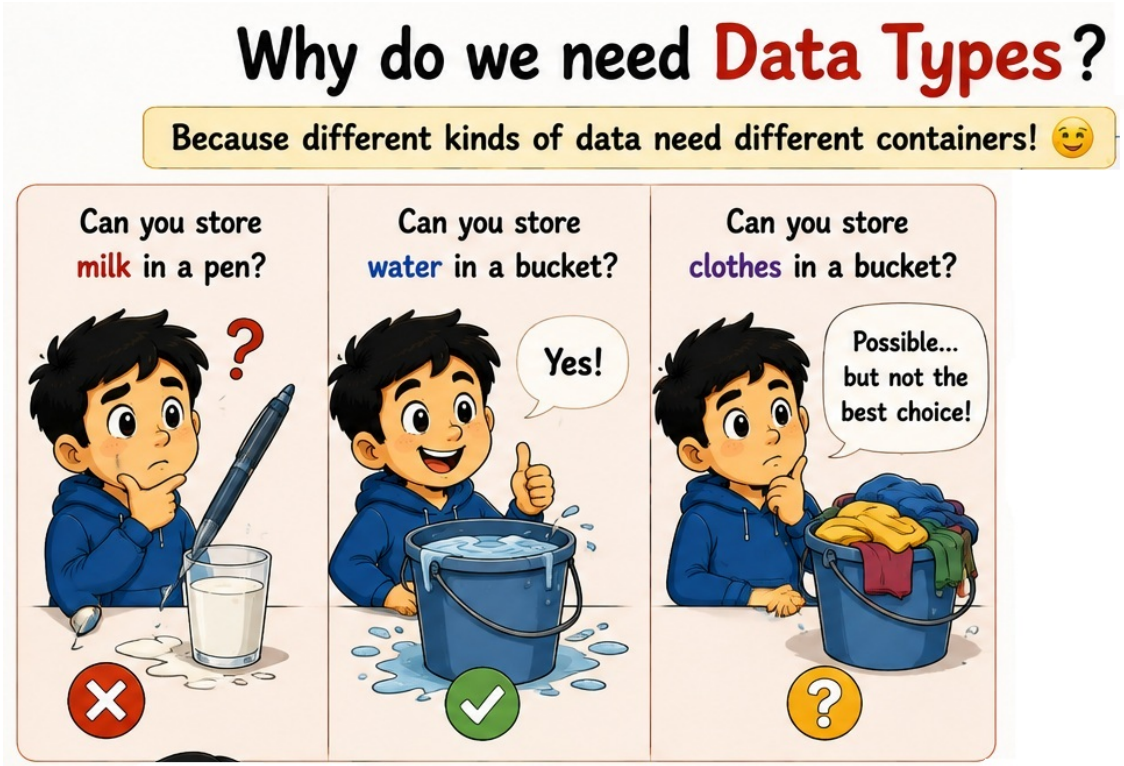
# Google Classroom

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad



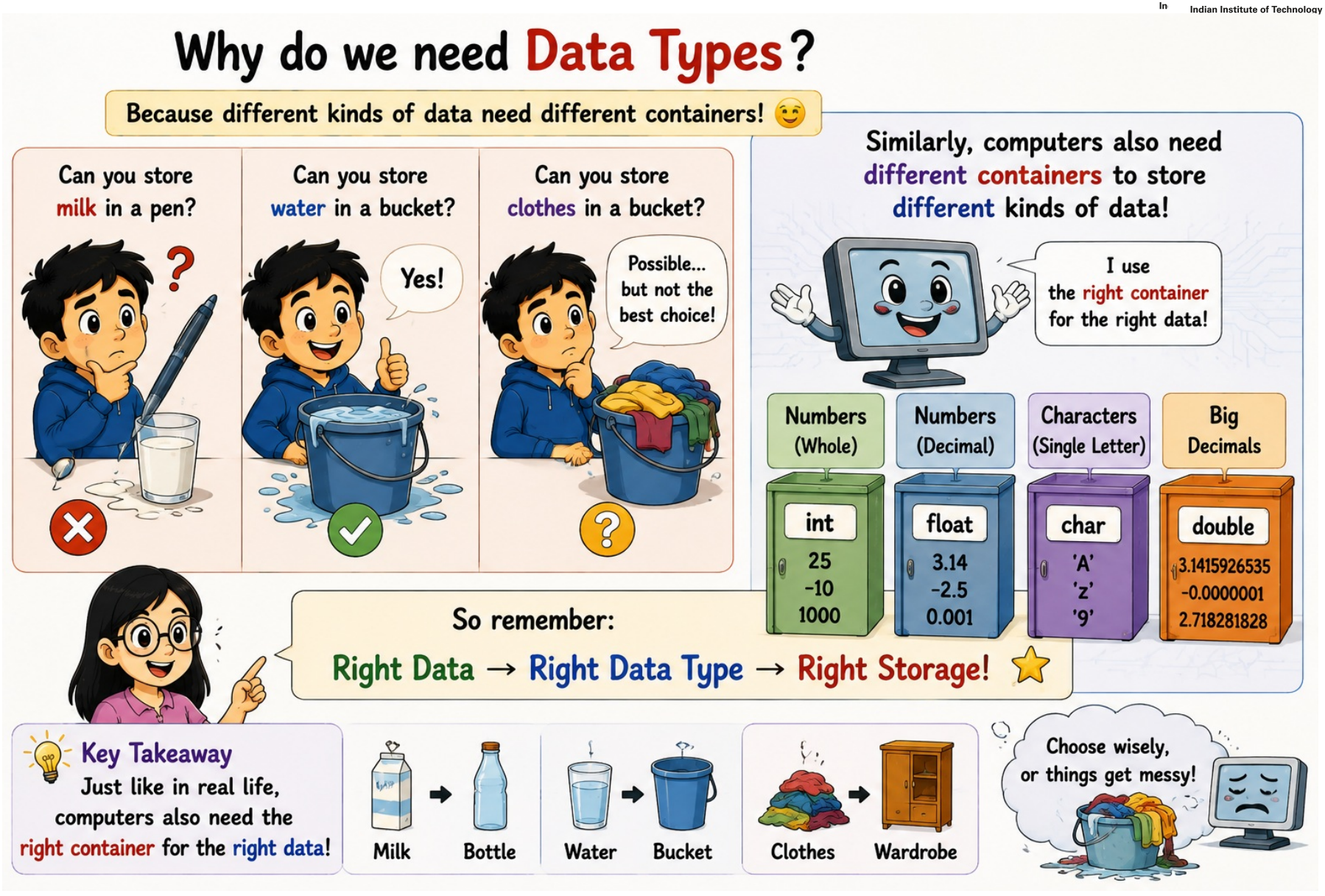
# Data Types




Why do we need Data Types?
Because different kinds of data need different containers!
Can you store milk in a pen?
Can you store water in a bucket?
Yes!
Can you store clothes in a bucket?
Possible... but not the best choice!

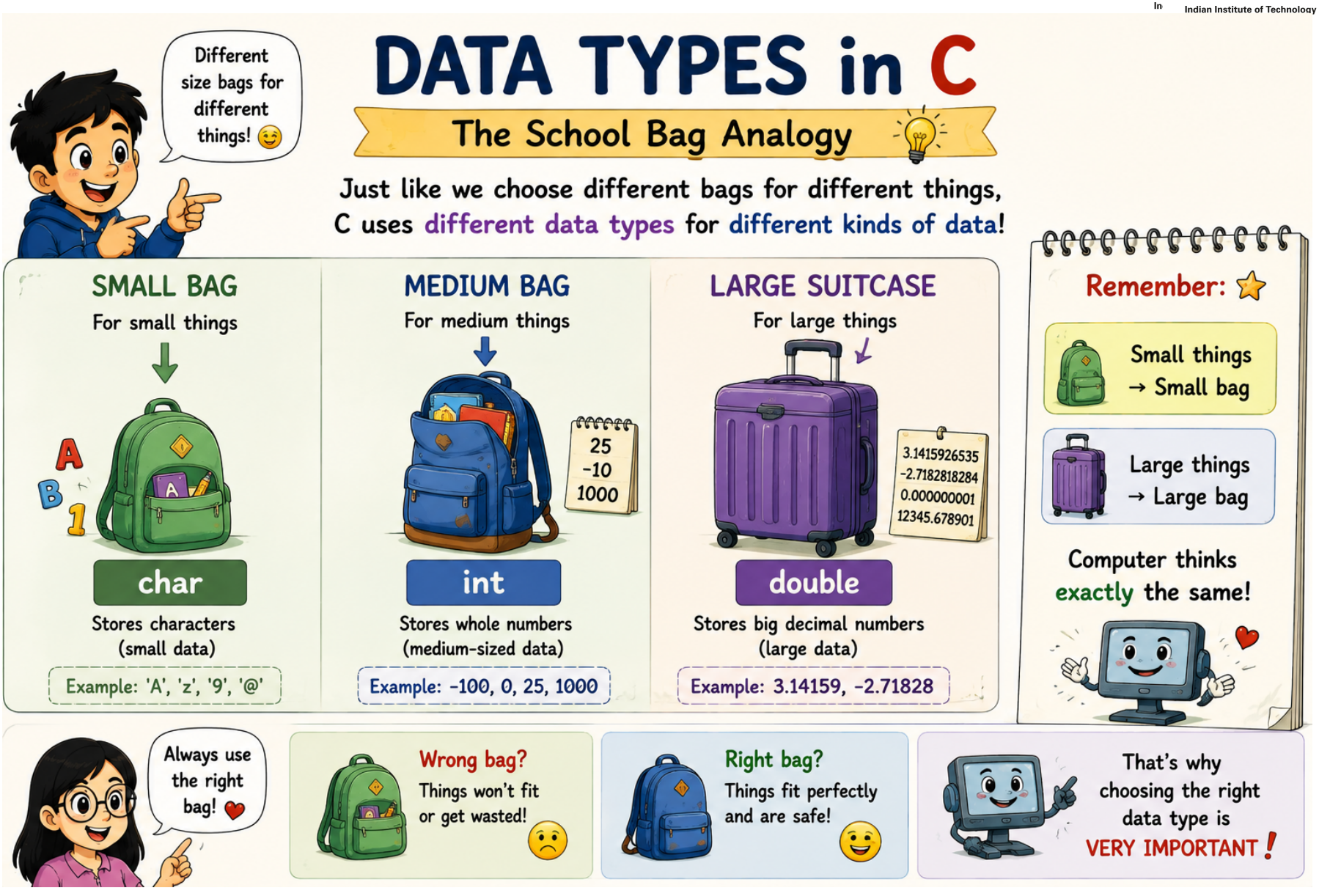
# Data Types




Why do we need Data Types?
Because different kinds of data need different containers!
Can you store milk in a pen?
Can you store water in a bucket?
Yes!
Can you store clothes in a bucket?
Possible... but not the best choice!
Similarly, computers also need different containers to store different kinds of data!
I use the right container for the right data!
Numbers (Whole)
int
25
-10
1000
Numbers (Decimal)
float
3.14
-2.5
0.001
Characters (Single Letter)
char
'A'
'z'
'9'
Big Decimals
double
3.1415926535
-0.0000001
2.718281828
So remember:
Right Data → Right Data Type → Right Storage!
Key Takeaway
Just like in real life, computers also need the right container for the right data!
Milk
Bottle
Water
Bucket
Clothes
Wardrobe
Choose wisely, or things get messy!

# Data Types

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad



# Data Types



| Data Type | Size | Example |
|---|---|---|
| `char` | 1 Byte | `'A'` |
| `short` | 2 Bytes | `120` |
| `int` | 4 Bytes | `25` |
| `long` | 4 Bytes | `100000` |
| `float` | 4 Bytes | `3.14` |
| `double` | 8 Bytes | `3.1415926535` |
| `pointer` | 8 Bytes | `&x` |

Table: Sizes of Basic Data Types on 64-bit Windows system using MinGW/MSYS2 GCC

# Declaring variables



- Must declare variables before use.
- Variable declaration:
  int n;
  float phi;
- int - integer data type
- float - floating-point data type

# Initializing variables

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

- Variables initialized via assignment operator:
  n = 3;
- Can also initialize at declaration:
  float phi = 1.6180339887;
- Can declare/initialize multiple variables at once:
  int a, b, c = 0, d = 4;
- Uninitialized, variable assumes a default value.
  - Local variables do NOT get a default value. Their value is indeterminate (garbage).
  - Global variables are automatically initialized to zero.

# Default value of variables

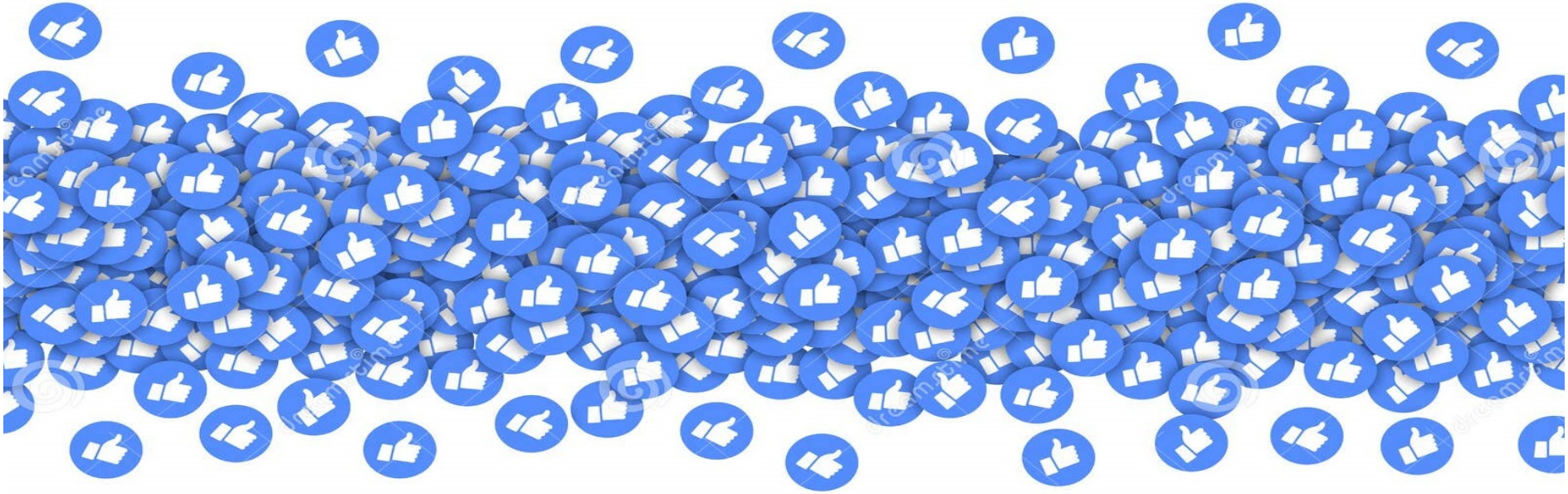
॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

- Uninitialized, variable assumes a default value.

```
1  #include <stdio.h>
2
3  int globalVar;
4
5  int main()
6  {
7      int localVar;
8
9      printf("globalVar = %d\n", globalVar);
10     printf("localVar  = %d\n", localVar);
11
12     return 0;
13 }
```

**Output**

```
global variable = 0
local variable  = 32767
```



**Thank you for your kind attention!!**